

Review

Climate-Resilient Microbial Biotechnology: A Perspective on Sustainable Agriculture

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Abstract: We designed this review to describe a compilation of studies to enlighten the concepts of plant–microbe interactions, adopted protocols in smart crop farming, and biodiversity to reaffirm sustainable agriculture. The ever-increasing use of agrochemicals to boost crop production has created health hazards to humans and the environment. Microbes can bring up the hidden strength of plants, augmenting disease resistance and yield, hereafter, crops could be grown without chemicals by harnessing microbes that live in/on plants and soil. This review summarizes an understanding of the functions and importance of indigenous microbial communities; host–microbial and microbial–microbial interactions; simplified experimentally controlled synthetic flora used to perform targeted operations; maintaining the molecular mechanisms; and microbial agent application technology. It also analyzes existing problems and forecasts prospects. The real advancement of microbiome engineering requires a large number of cycles to obtain the necessary ecological principles, precise manipulation of the microbiome, and predictable results. To advance this approach, interdisciplinary collaboration in the areas of experimentation, computation, automation, and applications is required. The road to microbiome engineering seems to be long; however, research and biotechnology provide a promising approach for proceeding with microbial engineering and address persistent social and environmental issues.

Keywords: microbiome; organic agriculture; environment; bio-engineering; biosensors



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1. Introduction

According to several reports and predictions, the global population will surpass nine billion by 2070 and the demand for staple food is expected to increase by at least 70% by 2050 [1–3]. Crop production to feed such a large population will undoubtedly have a tremendously adverse effect on the biosphere, posing a daunting task for the agriculture sector to solve. This problem is expected to worsen within the coming few years [4]. Enhanced agricultural production through the sustainability of the biosphere is essential in order to satisfy the increasing demands of the growing population [5]. In the face of current agricultural and environmental challenges, increasing agricultural productivity is of great significance for improving the global standard of living and enhancing the sustainability of food production [6]. The research investment in agricultural production has significantly promoted the increase in agricultural productivity, but there are large regional differences in the improvement of production levels [7]. Moreover, in the past century and a half, climate change caused by human activities has had a certain impact on agricultural production. However, most current studies focus on the impact of future

climate change on agriculture [8]. The impact of climate change caused by historical human activities on the agricultural sector has not yet been quantified.

In the context of returning to nature with scientific and technological progress and improving living standards, people began to search for solutions to reduce environmental pollution [9]. However, the focus on increasing crop productivity to accommodate the immediate food demands of the human population develops critical environmental vulnerabilities, including global warming, diminishing arable land, water scarcity, and excessive utilization of fertilizers and pesticides [10]. Despite this, fertilizers have made a tremendous transformation in increasing crop yields. According to the data provided by the International Food and Agriculture Organization (FAO), the contribution of fertilizers to agricultural input is 40~60% [11]. Excessive use of chemical fertilizers is directly responsible for the hardening of farmland soil and biological changes in soil flora, whereas it indirectly triggers a series of ecological concerns, food safety issues, social/economic losses, and several environmental problems [12].

Global warming, climate change, and environmental pollution enact plants with unique combinations of different abiotic and biotic stresses [13]. Although much is known about how plants acclimate to each of these individual stresses, little is known about how they respond to a combination of many of these stress factors occurring together, namely a multifactorial stress combination. Recent studies revealed that increasing the number of different co-occurring multifactorial stress factors causes a severe decline in plant growth and survival, as well as in the microbiome biodiversity that plants depend upon. This effect should serve as a dire warning to our society and prompt us to decisively act to reduce pollutants, fight global warming, and augment the tolerance of crops to multifactorial stress combinations.

Plants have developed several biochemical and structural defense mechanisms against biotic and abiotic stress to survive under extreme or harsh environmental conditions [14]. Plants and microbes have coexisted in the environment for a long time, and their interaction, whether synergistic (association of plants with endophytes and soil micro-organisms conferring stress resistance to biotic and abiotic factors and stimulating plant growth) or antagonistic (host–pathogen interactions conferring disease development), may have a crucial effect on sustainability in agriculture [15]. In the natural ecosystem, microorganisms mainly perform their functions and driving roles in the form of microbial communities, which maintain the structure, function and dynamic balance of the ecosystem at all levels [16]. Microbiome research has become the focus of global scientific and technological competition in plant disease prevention and control, industrial and agricultural production, and environmental management, etc. [17]. With recent progress in microbiome studies, we have some initial knowledge about the composition of microbial communities in various ecosystems, but we are still far from a comprehensive understanding of their community dynamics and functions.

Establishing an interaction between plants and microbes can provide several benefits: coexistence, reciprocity, and involvement, which improve plants' immunity [18]. Similarly, microbes such as bacteria and fungi also secrete a series of chemical compounds that can change the composition of root exudates [19]. The root systems usually provide anchoring and water transmission and fix root emissions, carbon dispersion, and micro-nutrients (amino acids, sugars, organic acids, and polysaccharides) into the soil to deal with disease-causing microbial neighbors, thereby encouraging microbial activity [20].

Current management strategies by prospecting plant–microbe interaction to mitigate soil contamination in agro-ecosystems derived from heavy metals, pesticides residues, emulsifiers, and surfactants, have been proposed [21,22]. Under the umbrella of agricultural modernization and sustainable development, the time has come to develop more eco-friendly microbial fertilizers to replace some chemical fertilizers [23]. As a new microbial preparation in biological fertilizers, the microbial agents (bio-agents) have many beneficial functions, such as increasing production, disease resistance, and fertilizer use efficiency [24]. With the rapid growth of the economy and society, people's awareness of ecological and environmental protection has immensely improved. Moreover, micro-organisms are widely

distributed in every corner of the natural world, and they are an essential part of the biogeochemical cycle.

In this case, the application of macrobiotic technology and its related products in the agricultural field will be helpful to solve these problems, whereas applying agricultural-associated micro-organisms and technologies in agriculture can help reduce the pollution of the environment and agricultural products caused by conventional chemical pesticides and fertilizers and help maintain the safety of the ecosystem [25]. At present, the application of microbes in agriculture has become a research hotspot. European and American countries have developed rapidly in researching and applying microbial fertilizers, biocontrol bacteria, and related agricultural microbial agents. They are leading the research and development of new products in this field. In recent years, the development of agricultural microbiological technology has been relatively rapid and has achieved good results in its application, but there are still many shortcomings [26,27].

Farmers are relying on the use of chemicals to boost agriculture production and yields which creates health hazards to human and environmental issues. Keeping in view, United Nations adopted a protocols on agro farming and biodiversity to reaffirm the sustainable agriculture, which contributes more to biodiversity conservation than conventional agriculture, but outcomes are reduced and increased cost of production and low yields [28]. Plant microbiome hold the ability to increase sustainability and disease resistance in plants [29] and they can change the face of Agriculture farming [30]. Recent research reveals that microbes can bring up the hidden strength of plants [31]. Apples grown without chemicals and fertilizers [32], Rice that are more resistant against diseases and pests [33], Tomatoes cultivated with Endophytes grown taller as compared grown without endophytes [34,35]. Hence, crops could be grown without chemicals by harnessing microbiome. Plant microbiome mainly consists of soil microbiota and Endophytes, which are microorganisms that live inside the plants (e.g., Fungi bacteria, Archaea, etc.). Endophytes are defined as “Microorganisms within plants that do not cause disease” [36]. These are mutualists or symbiotic partners that benefit the plants in multiple ways, i.e., in nutrient acquisition, stress tolerance, Pathogen resistance pollution degradation, and plant growth promotion [37]. The Endophytes study describes the significance of indigenous plant microbes to host phenotypes such as growth, vigor, and health to understand the functions of microbial individuals and communities, and to explore host–microbial and microbial–microbial interactions and maintenance of the molecular mechanism of microbial communities [38].

Several references for the application of climate-resilient microbes for the development of enhanced ecosystem services and resilient crops consider tangible and logical proofs for the plants thriving in harsh environments that exhibit a symbiotic relationship with microbes as endophytes or phytomicrobiome, helping them to withstand extreme environments [39,40]. However, there are still several scientific ambiguities to address for a better understanding of the mechanisms and evidence for microorganisms’ response, adaptation, and evolution in environments pretentious to extreme events during climate change [41].

Hereafter, to eliminate some scientific haziness established in the description specified earlier, this review covers the theoretical and practical research focused on applications of microbes to augment the inner strength of plants enabling them to thrive under wide-ranging environmental conditions subjected to climate change. There is particular emphasis on designing climate-resilient crops in consortium with resilient microbes that are capable to adjust swiftly to the current agroecosystem. This phenomenon is quite a multifaceted series of interactions that are supposed to require comprehensive modeling for refining predicted outcomes.

To promote the healthy development of agricultural microbiotechnology and related industries and promote agricultural modernization to the next level, it is necessary to further strengthen the basic and applied research in this field. Hence, the development and application of microbial agents are described in this review. We focused on the recent state-of-the-art techniques and concepts for the consortium of particular plant microbiomes and discussed the principles and best practices for climate-smart agriculture. Meanwhile, a

simplified method is discussed that explores the experimentally controllable synthetic flora used to perform targeted operations in a limited biological system to study plant microbial communities and to simplify the complexity of in situ studies.

2. Methodology

Climate-smart agriculture with sustainability is envisioned to augment agroecosystems supported by extreme-resilient organisms that are able to withstand during harsh climate events. To evaluate this argument, the present review is designed to explore the integration of various techniques described in previous literature and elaborate on the frontiers of microbial biotechnology.

All the records (papers, reviews, reports) retrieved in the initial instance were curtailed for relevance according to the subject, research type, and focus of the present study. We retrieved most of the literature from the most common, easily accessible and authenticated databases such as Scopus/ScienceDirect, Web of Science (WOS), Springer, MDPI, Wiley, Tandfonline, directory of open access journals (DOAJ), and Google scholar (grey literature). During the first step, we tried to retrieve the latest literature from the last five years (2017–2022) then expanded to 2010. The retrieved literature was screened on the basis of relevance with the title of the present review and keywords. We segregated the eligible literature based on time and relevance, which is cited later in this review, and excluded the remaining records. The number of publications during the timescale (2010–2021) falling into different subject categories is presented in Figure 1). The current review also demonstrates the latest information about microbes in different sectors of agriculture biotechnology and mainly used as bio-fertilizers, bio-pesticides, and many publications found in environmental sciences. In recent years, it has been observed that the utilization of microbes has significantly increased over the decade of research (Figure 1).

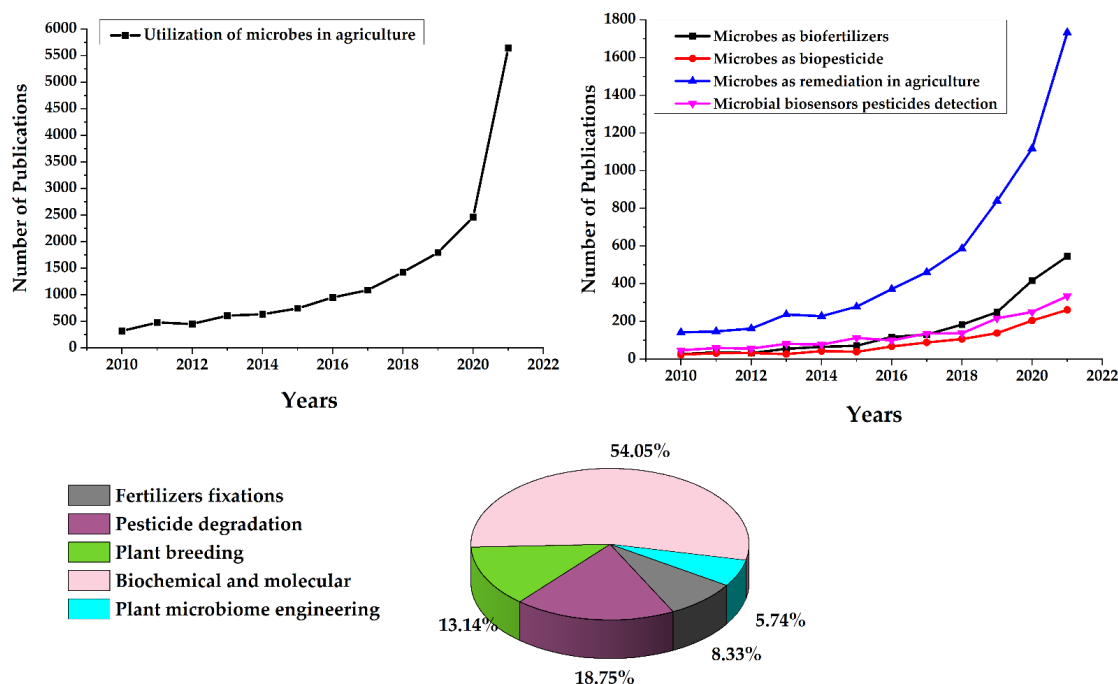


Figure 1. Recent development status of the utilization of various microbes in agriculture and publication percentage in different journals. These data have been taken from Science Direct Scopus journals.

The eligible records were retrieved and exported into Endnote software (references managing tool). All the identical and specious records were excluded as described in Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA 2020) [42], and after the screening, the bibliography was evaluated according to the subject, retrieved sources and types of literature, and the publication year (Figure 2).

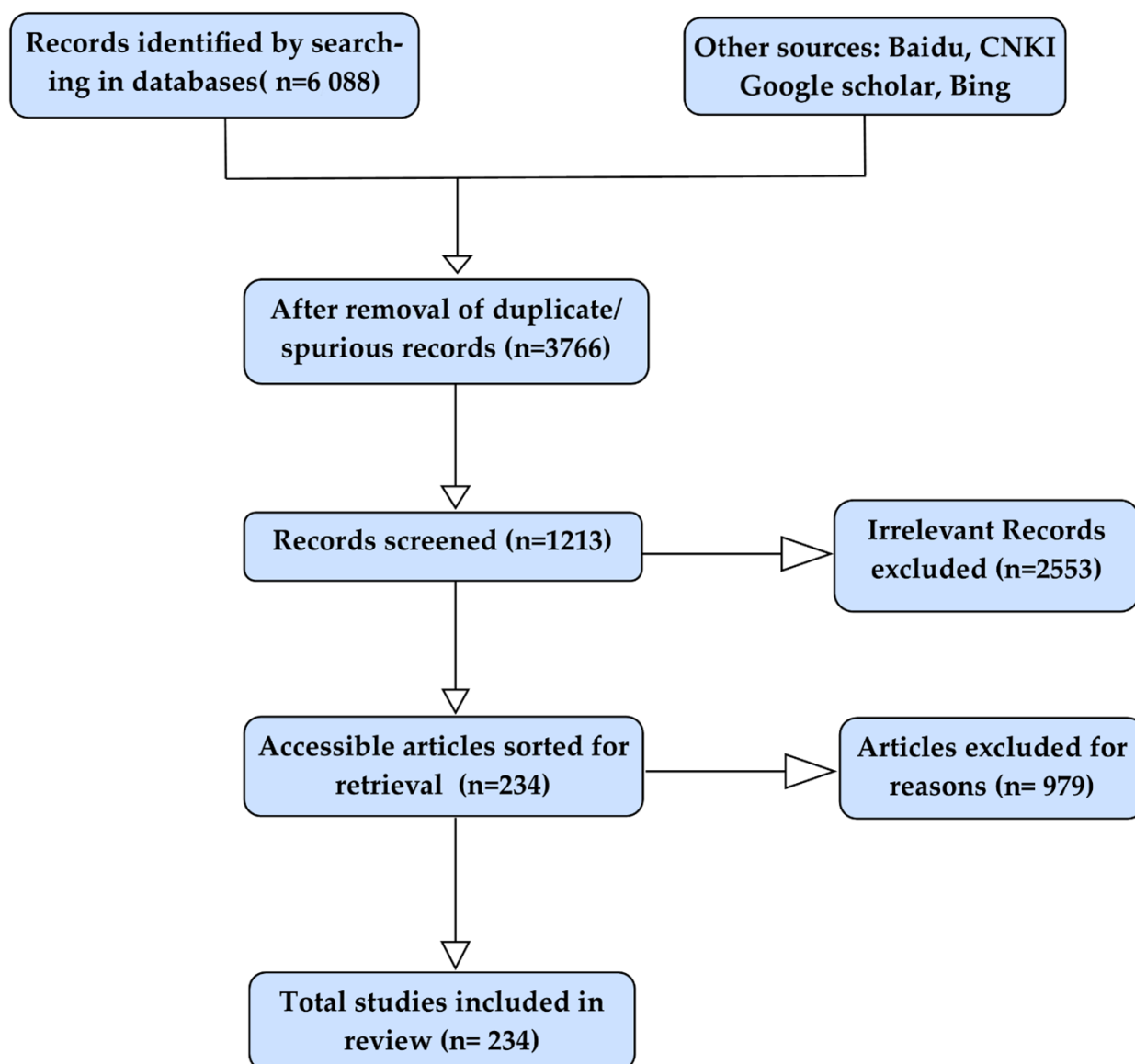


Figure 2. PRISMA chart of current research sources.

3. Climate Change

Climate change is a subject of growing interest in the twenty-first century because it threatens human well-being, agricultural production, forestry, biodiversity, and environmental health. Changing climate, the steady increase in CO₂ levels in the atmosphere, and rising temperatures often have a considerable adverse effect on ecological processes [43]. During the mid-twentieth century, micro-biotechnology facilitated the agriculture sector and subsequently sustained severe ecological consequences, leading to global pollution, adverse climate changes, and biodiversity extinction [44].

3.1. Harnessing Resilient Microbes for Climate Change

Climate and microorganisms are collectively responsible for driving plant growth, which ultimately have significant influences on crop yield, quality, quantity, and sustainable food production system [45,46]. Microbes play an integral role in modulating the cycles of various elements such as carbon, nitrogen, and oxygen [47]. Climate change affects the functions of microbial communities as a whole. The situation is further complicated by indiscriminately using agro-pesticides, fertilizers, and plant nutrients with a high salt index. Biotechnological interventions have helped to develop climate-resilient agriculture [48]. A

“next-generation Green Revolution” is required to achieve future food security. Radical new concepts and approaches are needed to achieve more sustainable development of agriculture [49].

Microorganisms thriving in diversified and extremely harsh habitations possess several physiochemical mechanisms to adapt extreme environments, such as temperature fluctuation, salt, pH, radioactivity, pressure, lack of oxygen, and different chemicals. At the same time, some poly-extremophilic organisms have abilities to tolerate and grow under multiple stress and extreme environmental situations [50]. These qualities make them suitable to harnessing and designing synthetic microbial communities for improved crop resiliency [51], practically applied in climate-smart agriculture systems either directly as plant microbiome, endophytes, and signal exchange between plants and microbes [52] or indirect applications such as production of hormones (gibberellins, auxins, and cytokinins), extremozymes (cellulose, protease, xylanase), and various fields of the agriculture industry for bioremediation, and biodegradation [53]. Furthermore, the application of these climate-resilient microbes is also ideal in food processing industries such as fermentation, and the production of vitamins and flavors. The possibility of importing the biological traits of these extreme microorganisms to augment plants through metagenomics or CRISPR genome editing has many potential applications, including drought/salt tolerance, diseases/pest resistance, organic matter decomposition, heavy metal removal, nutrient cycling, climate resilience and improved growth and vigor of crops [54].

3.2. Climate-Smart Agriculture

Climate-smart agriculture (CSA) is an integrated agronomical approach that emerged as a framework for transforming and reorienting farmland, livestock, forestry, and inland fishery that interlink the challenges of food security and sustainable development goals during climate change events [55]. Microbes play a dynamic role in various bio-ecological series of actions in agroecosystems such as decomposition of crop and animal residues, micronutrient recycling, nitrogen fixation, phosphate hydrolyzing, nutrient-acquisition, and the benefits of probiotics [53]. The microbial constituents of crops and soils are well recognized and are increasingly being utilized to manage the adverse impact of climate change [56]. The application of co-adapted, mutualistic plant-microbial interactions (endophytic and rhizosphere) collaboration with hybridization practices and biotechnology potentially results in an agriculture approach characterized as constructed microbiomes [57]. A designed microorganism's community strategy is a microbial combination utilized to construct and produce mutualistic microbial communities via exploiting ecological and evolutionary characteristics [58].

Much uncertainty still remains concerning the resilience of plants, soils, and associated microbes to climate change. Intensive efforts are underway to improve crop yields with lower input requirements and to enhance sustainability through improved biotic and abiotic stress tolerance traits [59]. Plant growth-promoting microbes function as bacterial inoculants and contribute to the enhancement of agronomic efficacy by lowering production costs and environmental pollution [60]. The appropriate management and exploitation of beneficial microbes ranges from endophytic fungi to plant growth-promoting bacteria, and beneficial microbial functions such as bio-fertilizer and bio-pesticide are a sustainable method in producing climate-resilient crops [61].

Metagenomics' theoretical and applied tenacities provide a clear vision of eco-biological diversity and conservation characteristics of microorganisms, providing comprehensive and valued approaches for augmented plant species to withstand climate change and global warming [62]. Because of climate change, agriculture production from arid and semiarid areas is declining due to salt stress and lack of irrigation, whereas the conventional agriculture system already does not support sustainable agriculture [63]. The pressure on arable lands was increasing at the world level due to the ever-increasing nutrition demand from the escalating global population and estimated irrigation water limitations as expected from climate change [64]. Maximum utilization of degraded soils with best

practices requires time, and it can only be achieved through the symbiosis of the ideal microbiome with plants. The use of symbiotic microbes isolated from plants thriving in extreme environments appears to be a powerful tool for successful tolerance in vulnerable plant species [65]. Plants living in extreme environments show symbiosis with microbes and help the plant to survive in stressful environments [54]. The activity, diversity, and composition of the phyto-microbiome are often regulated by the signal exchange between plants and microbes [52]. Recently, several mutualistic microorganisms are being utilized with the intention of augmenting crop production, particularly in stressful environments [66]. On the other hand, some researchers investigated the effects of a solo symbiotic partner, which can also involve a wide range of functional and biochemical properties. In this regard, the full understanding of the nature of every plant microbiome relationship for comprehensive remuneration of potent symbiosis can benefit through consortium approaches [67].

Extremophiles are the microorganisms thriving in extreme environments apparent with harsh conditions such as thermophilic (high-temperature), psychrophilic (low temperature), halophilic (hypersaline environment), acidophilic (low pH), alkaliphilic (high pH), piezophilic (high pressure/deep sea), radiophilic (high radiation), xerophilic (no water), anaerobes (absence of oxygen), and snottite (in caves), and where normal life does not exist [68]. Our planet possesses various combinations of different ecological environments, which range from mild to extremely harsh, based on human perception, but microbial nature is ubiquitous and present in all places. Extremophilic microbes require particular mechanisms to flourish in such severe situations. Their amazing physiognomies can be utilized to combat such complications faced by the agriculture sector in response to climate change [46]. Microorganisms adapt to extreme environments by virtue of enzymes, secondary metabolites, and other bioactive compounds, which can be an attractive resource with great potential for use in agriculture biotechnology [69]. Naturally, soil micro-fauna decomposes plant debris, carries out nutrient recycling to sustain soil fertility, and consequently plays a significant role in biogeochemistry; the organisms that thrive in extremely low temperatures perform the same role. Recent research focuses on cold-adapting microbes, their ecology, enzyme activities, gene expression, metagenomic and biotechnology applications, and exploitation in the agriculture sector [70].

Metabolic engineering and synthetic biology are expected to accelerate crop improvement significantly. A defining aspect of both fields is the design/build/test/learn-to cycle, or the use of iterative rounds of testing modifications to refine hypotheses and develop the best solutions [71]. The eventual purpose of a rational microbiome design is to construct a mechanism that allows designers to enhance, eliminate, or alter specific functions and phenotypes directly in situ under ideal operating conditions [71]. Further development of the combination of in situ Metagenomic design and CRISPR-CAS gene-editing technology will enable the exact employment of the metabolic network of in situ micro-organisms, efficiently mingling self-assembled and synthetic microbial groups [72].

Endophytic microbes are important players in making agriculture resilient to climate change. One of the management strategies for dealing with the adverse effects of abiotic and biotic stresses resulting from climate change is the application of extremophilic endophytic microorganisms [73]. Synthetic ideal endophytes are not only a substitute for environmental research, but rather a molecular basis that complements the interpretation of natural phenomena and can be transferred to a controlled experimental system for further research [74]. Conversely, the mechanisms and theories obtained can be verified in the greenhouse or in vitro experiments. The reference model system is of great help in establishing the causal relationship between endophytes and plants. In short, it is necessary to deepen the understanding of the basic principles of synthetic endophytes and apply this knowledge to the agricultural industry and crop production [75].

4. Plant Pathogens, Biocontrol Strategies, and Metabolomic Applications

Several soil-borne plant pathogens (such as *Burkholderia* sp., *Xanthomonas* sp., *Penicillium* sp., *Fusarium* sp., *Pseudomonas* sp., *Rhizoctonia* sp., and *Alternaria* sp.) produce bioactive metabo-

lites (fumonisin, toxoflavin, albicidin, citrinin, RStoxin, coronatine, and alternariol, respectively), which damage the quality and quantity of cultivated products which are harmful to livestock and human health. Certain phytopathogens are mainly related to fusariosis, cereal mildews, and smuts/spots and can have a massive influence on many people across the globe, particularly those who depend on cereals (such as maize, wheat, etc.) and barley as a staple nutritional supply [76–79]. Numerous pest and disease management strategies to mitigate the emergence are dependent on the employment of resistant cultivars and chemical pesticides [80]. In recent years, the application of microbes-derived bio-pesticides as safe and environmentally friendly approaches has gained special attention, particularly in organic farming [81,82]. Notwithstanding this, neither of the chemical-derived techniques has comprehensively eradicated these noxious phytopathogenic fungi.

On the other hand, their residues exceptionally prevail in foodweb and the environment, particularly in developed countries where use is abundant [80,82]. However, various conventional preventative measures are already in practice to avoid pre-harvest yield losses or post-harvest/storage crop losses (e.g., sun drying, air drying, and phytochemical applications). In addition to this, several contaminating and toxicogenic micro-organisms are deemed responsible for quality losses in crops since frequent reports of human and animal poisoning from such toxins still occur and are evident [77–79,83].

4.1. Plant Pathogens and Control Mechanisms

It is a well-known fact that plants and microbes are two of the most important and integral living creatures on earth. These creatures are abundant and can inhabit and tolerate an extreme environment and stress with mutualism. However, the interaction between these two groups has been reported for a long time [84]. Recently, studies related to plant–microbe interactions and molecular analysis have seen an incredible advancement. Awareness about mechanisms of the plant–microbe interaction, which can either be disease or resistance, is the primary basis used to categorize them as friendly or hostile. A non-phytopathogenic root-colonizing bacteria, also known as Plant Growth-Promoting Rhizobacteria (PGPR), is commonly utilized as a bio-fertilizer and/or biological control agent, a harmless and ecologically favorable alternative to chemical options [85].

In general, suitable and compatible bio-fertilizers with excellent development and biocontrol activities combined with disease prediction systems may combat serious crop infirmities, thereby seeking sustainability in agriculture [86,87]. However, researchers isolated *Pseudomonas* and *Bacillus* bacteria from wheat grain varieties planted in calcareous soil. The isolated bacteria containing plant growth-promoting (PGP) components could be investigated *in vitro* for additional PGP characteristics, including indole synthetic chemistry and biological control of plant pathogens, that could help promote plant growth [88]. The current review suggests that multi-trait rhizobacteria, which are active components of the arylsulphatase-producing rhizobium group, can increase, encourage, and potentiate the natural positive microbe-host association in agro-ecosystems. Biocontrol of plant pests and diseases seems to become the tremendous preference for developing cost-effective, eco-friendly, and effective control practices to secure crops [89,90]. Biological control strategies, particularly those that are microbe-based, are increasingly rapidly regarded as potential tools against plant diseases for sustainable farming practices.

Hence, there are numerous biocontrol methods available. The future development of particular methods will require thorough knowledge about the complex interactions among plants, the environment, and pathogens. The biocontrol strategies might not be efficient when plants are already showing disease symptoms [91]. The main objective of biological control studies seems to diminish the dependency on agro-chemicals and the concerns for human health and the environment.

Deciphering the potential of knowledge about the defense mechanisms in plants since evolution and identification of the biological drivers involved is a big challenge. The plant defense systems can be segregated into three basic mechanisms: prevention (antixenosis), confrontation (antibiosis), and tolerance (a recompense stratagem to diminish the harmful

effects of the pathogen). Plant defensive systems are passive defenses comprising non-host resistance, physical and chemical tolerance, immediate proactive defensive systems, and prolonged active defensive capabilities. Modifications in membrane functionality, the first oxidative burst, cell wall restoration (Figure 3), hypersensitive response (HR), programmed cell death (PCD), and all components of phytoalexins are accelerated to active defenses [92,93]. Interrupted functional reasons contain pathogen restraint and injury repair, systemic acquired resistance (SAR), and pathogenesis-related (PR) gene expression. Salicylic acid (SA), which is crucial for defense over bio-trophic infections and SAR, as well as jasmonic acid (JA) and ethylene (ET), are engaged during necrotrophic pathogen defense as well as advantageous plant–microbe interactions, such as priming and induced systemic resistance (ISR) [94].

Potential biocontrol microbes can easily be grown by culturing them using a suitable substrate as a medium. These substrates constitute the nutritional compositions that microbes require for cell multiplication, development, metabolism, and function. A study by Yang Bai showed that a significant proportion of related micro-organisms could be cultured utilizing systematic bacterial extraction procedures. This may be beneficial in attracting advantageous microbial populations within the existing soil microbiota and importing and conserving beneficial microbes for biological control of plant pathogens through supplying exemplary substrates as the growth medium. Helpful microbes, particularly bacteria, are adaptable to various habitats and circumstances due to high metabolic versatility such as temperature, salinity, acidic, etc. [95].

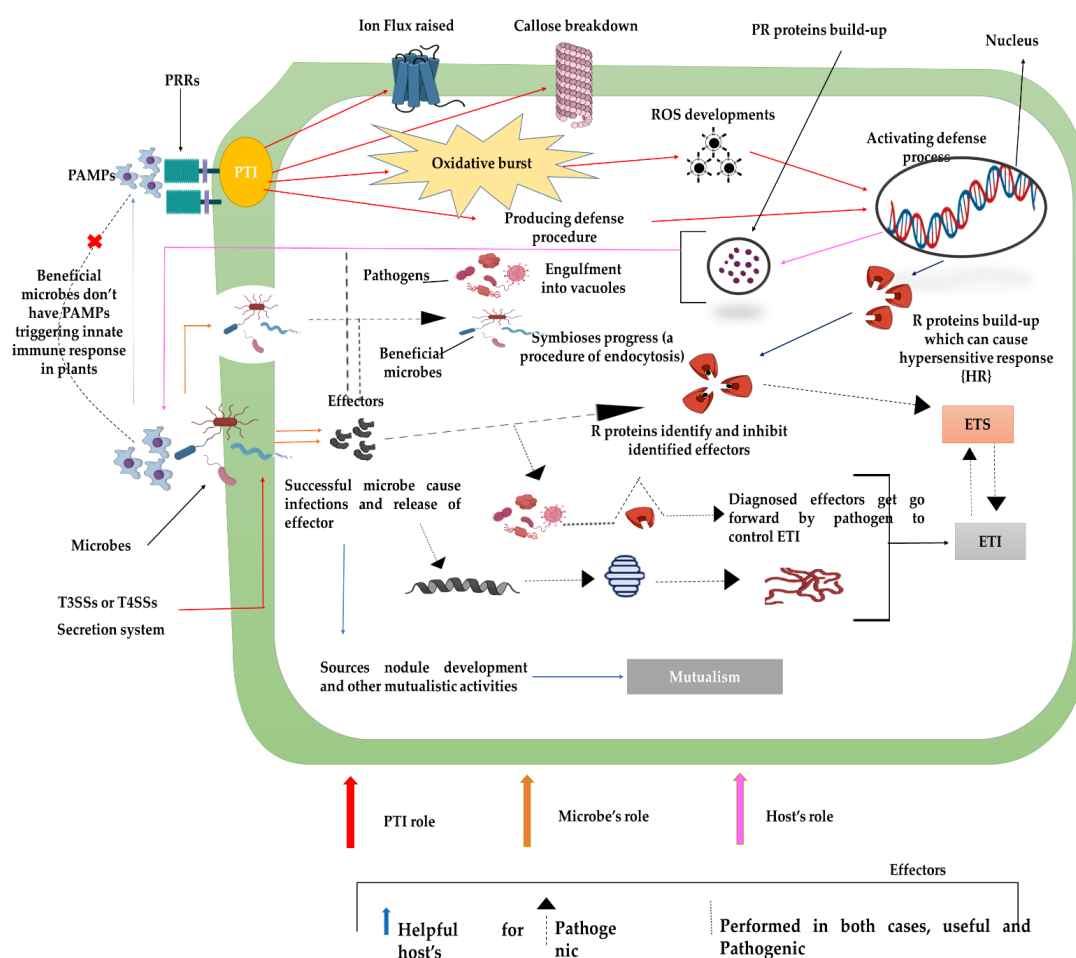


Figure 3. This figure was adopted by [96] with minor modifications. This illustration shows the plant–microbe interactions represented in mutualistic and pathogenic activities.

4.2. Plant–Microbes and Metabolome Variations

Metabolomic techniques had already been successfully utilized to characterize and diagnose crop diseases and epidemiologic agents effectively [97–99]. However, its application in modern agricultural research focused on tripartite (plants–pathogenic microbes–beneficial microbes) interactions seems very limited. Since novel methodologies for reducing food-borne diseases, promoting crop yields, and advancing agro-international supply chains are actively explored worldwide, an initiative to understand the tripartite microbes-derived bio-pesticides is important. The symbiotic influence of arbuscular mycorrhizal fungus (AMF) and plant growth-promoting rhizobacteria (PGPR) consortia upon the metabolism of *Triticum durum* Desf. (Durum wheat) had been assessed utilizing the GC-TOF-MS and HILIC-Q-TOFMS (hydrophilic interactions chromatographic time-of-flight mass spectrometer) [100].

Metabolomics techniques such as HS-SPME-GC-MS and LC-HRMS were used to analyze the dynamics of volatile and non-volatile metabolites following fungal cohabitation. The researchers examined the metabolomes of two grapevine infectious agents, *Eutypa lata* and *Botryosphaeria obtusa* (in co-culture). They assessed whether individuals responded to commercially available 2-nonanone antimicrobial metabolites, the sources of which were unspecified. Within one week, there was complete suppression of the two grapevine pathogenic microbes. The additional most important volatile and non-volatile metabolites identified in the research included decane, an unknown substance from the sesquiterpene class, and O-methylmellein [101]. Sevastos et al. (2018) reported that utilizing proton nuclear magnetic resonance (^1H NMR) metabolomics is a methodology that can be applied to characterize the metabolic interruptions influenced in the metabolome of non-transgenic *Fusarium graminearum*; four carbendazim-resistant *F. graminearum* isolates and an untreated *F. graminearum* strain were identified (within a week of treatment with the synthesized fungicide carbendazim). The researchers determined a positive correlation between a few metabolite contents (up-regulation: L-serine, L-phenylalanine, D-glucose, L-glutamate, L-methionine, pyroglutamate, and citrate; adversely regulated: D-myoinositol, threonine, malate, and L-sucrose) identified in wild and resistant *F. graminearum* cultures that could be subsequently applied in the future [102].

Bucher et al. (2017) also applied GC-MS and LC-MS metabolomics to investigate the effects of Lr34 (a multi-phytopathogen resistant conferred gene) on field-grown and transgenic greenhouse-cultivated barley, rice, and wheat. The plants came back positive against powdery mildew and rust diseases. Whereas most of the secondary metabolites identified in the research were recognizable, nine primary metabolites and 16 lipids were interrupted in barley plants. In barley plants, glucose and fructose contents were maximized, whereas dehydro-ascorbate rates were reduced [103]. In general, 84 primary metabolites were detected from the transgenic plant's samples containing amino acid compounds, sugars, organic acids, polyols, and various lipid groups [104].

Aliferis et al. (2014) utilized an Orbitrap MS (directly infusing) and GC-MS technique to track the response of soybeans to *Rhizoctonia solani* infection throughout time. The method also included creating a complete soybean metabolite library, which aided in identifying metabolites and analyzing data produced throughout the research. Inside the pool of metabolites examined, biomolecules such as phytoalexins, coumarins, and flavonoids reported improved soybean defensive characteristics towards biotic stress [105]. In addition to the procedures discovered, the current review suggests that this method can be applied to cereal grain studies in the future. Furthermore, 1-methyltryptophan was found as a novel biomarker associated with the phytopathogen-infected tomato metabolome, and the hexanoic acid stimulated tomatoes' metabolic profile. The research revealed how metabolites produced from plants respond to both biotic and abiotic interruptions as they are source-dependent [100].

The present review also recommends that tripartite plant–microbe symbiotic metagenomics may be complex due to the diversification of biological molecules and the large data obtained. New progress in metabolomics research methods (non-targeted and semi-targeted), equipment, and chemometrics/bio-informatics have also resulted in quicker,

simpler, and more observable data acquisition. We also anticipate a significant rise in the discovery and use of metabolite biomarkers in managed and semi-controlled planting methods. Food instability and several other concerns that growers experience in disease-prone areas of the globe may be minimized if crop protection strategies are effectively applied.

5. Plant–Microbes Symbiosis: Bio-Fertilizer

As a kind of bio-fertilizer, plant–microbes symbiosis plays a vital role in promoting plant growth. Improved plant nutrient utilization efficiency plays an excellent role in the sustainability of the agriculture sector. During a symbiotic interaction, bacteria utilize inactive nitrogen in the adjacent environment to transform nitrogen into a form that plants can easily absorb (ammonium and nitrate) and continue to receive carbon sources depending on the host plant [106]. Based on analysis, the Plant Growth-Promoting Rhizobacteria (PGPR) groups such as *Azorhizobium* and *Sinorhizobium* genera are responsible for nearly 65% of the entire nitrogen delivery agri-crops globally [107,108].

Soil bacteria constitute the most effective bio-fertilizer, in particular bacterium species such as *Mesorhizobium*, *Sinorhizobium*, and *Azorhizobium*. The nitrogen fixation procedure is primarily regulated by two gene clusters, viz, *nif* and *nod*. Various bacteria have already had chromosomal or symbiosis plasmids (*sym*) genes identified, which resemble the rhizobial T3SSs (Figure 3) that can participate in the release of specified “symbiosis developing proteins” [109,110]. Several plant-beneficial bacterium species, particularly *Pseudomonas fluorescens*, have been extracted utilizing this technology. Therefore, a wide range of bio-fertilizers depending on symbiotic interactions among plants and microbes are used extensively in farming as friendly alternatives to chemical fertilizers [111]. Further investigation is required to develop in vitro synergistic formulations of microbes under various stress circumstances (soil pH, availability of organic acids, stressors, carbon sources accessibility) to improve soil fertility and efficiency of the bio fertilization procedure (Figure 4).

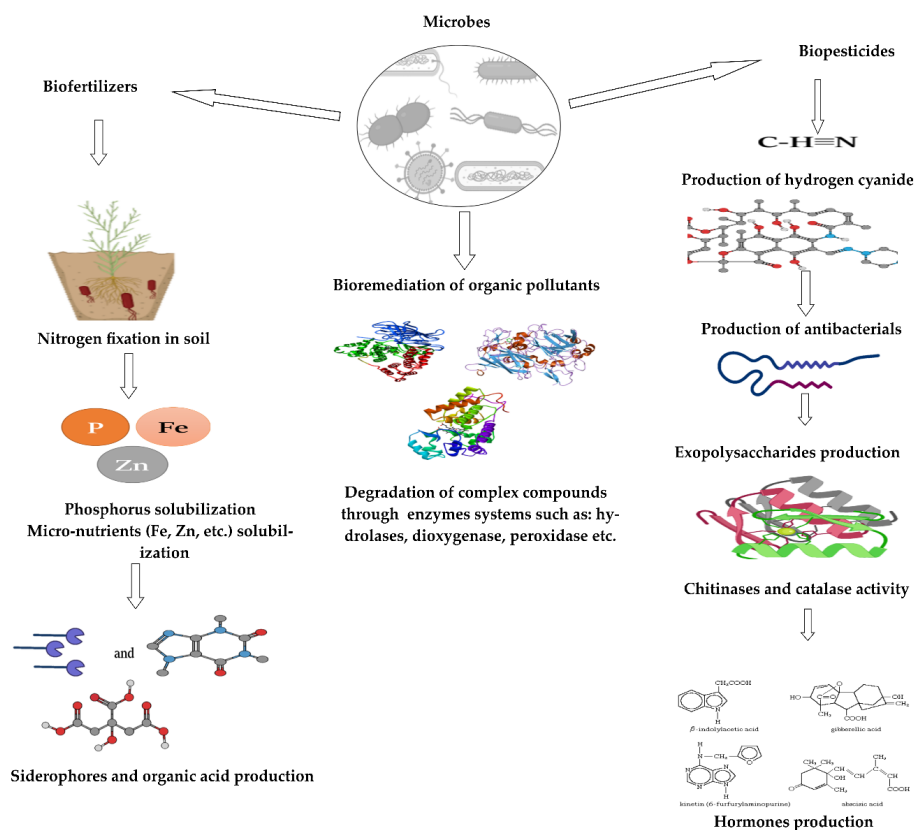


Figure 4. This diagram briefly discusses several applications of microbes in the agriculture field.

Application of Bio-Fertilizers to Control Replanting Disorders

In modern agriculture, replanting (re-establishing plants in soil where the same species was grown previously) diseases/disorders have become more prevalent. Using bio-fertilizers with beneficial microbes helps to prevent diseases and provides a potential approach to replanting disease management [112]. Therefore, the impact of dual replanting disease and bio-fertilizer amendment on crops' microbiota formation within leaves and roots, and related to crop production and quality, remained unclear. A recent study by Wu et al. (2021) indicates that bio-fertilizers can control replanting disease and pathogen intensity while also promoting supporting micro-organisms, including *Pseudomonas*, *Paenibacillus*, *Bradyrhizobium*, and *Streptomyces*. In addition, bio-fertilizers have a positive relationship in the co-occurrence and can increase the ecological commensalism or mutualism of the microbial community [113]. Another study by Dong et al. (2019) revealed that bio-fertilizers have the potential to decrease the existence of the *Panax* root rot disease of *Panax ginseng*. Furthermore, bio-fertilizers can benefit plant growth and yield by increasing the presence of highly relevant microbial species and reducing the rate of foreseeably deleterious microbial strains, which is an excellent step towards achieving sustainability in the future [114]. On the other hand, our review suggests that such kinds of applied methods ensure the safety of the medical plant due to the usage of medicinal plants in the human health sector.

6. Microbiome Engineering

6.1. Modification in Soil

Soil modification relates to organic and inorganic soil amendments or farming technologies utilized by growers to increase productivity and impact and influence plant-microbe interactions [115]. Inorganic modifications such as vermiculite, lime, perlite, and sand have been utilized to decrease the effects of soil salinity and acidity on crops [116,117]. They may also contribute to higher plant biomass due to improved root exudates and a more extensive, more active microbiome, including N-deficient soils. Inorganic supplements applied in tandem alongside organic additions enhanced crop production by 30%, compared to 8% using organic fertilizers individually [117].

Organic modifications constitute the injection of various compounds into the soil, which serves as sources of nutrition for heterotrophic bacteria and fungi, including invertebrates that can be advantageous for plants [116]. They provide an additional advantage of supplying depleted soil fertility and decreasing agricultural pollution. The continuous use of conventional organic fertilizers improves microbial activity and diversity depending on the amendments. The excellent example is understandable, having been utilized for many decades [118]. However, it is abundant in nitrogen, phosphorus, and organic material, affecting the soil microbiota and physical and chemical characteristics to improve agricultural production [119].

The combination of sheep dung stimulated microbial development and reduced toxic metals in alfalfa plants by diminishing lead, cadmium, and zinc concentrations, leading to tremendously increased crop production. Since the mid-nineteenth century, bones have been widely exploited as a sustainable phosphorus supply. Soil microbes, including *Bacillus megaterium* var. *phosphaticum* and *Bacillus mucilaginosus*, were extensively acknowledged for their capability to solubilize the phosphorus existing in the bone as well as render it approachable for plant intake without delivering organic acids (Figure 3) [119]. Crop remnants contain enormous amounts of organic carbon and other nutritional trash that may be valorized through recycling or vermicomposting to develop bio-stimulants that promote soil stability [118]. Simultaneously, end-products of vermicomposting are environmentally favorable and nutrient-rich and may be applied as soil conditioners compounds to support plant growth [120].

Several bio-stimulants (such as algae extracts, amino acids, humic-fulvic acids, and hormones) are supposed to accentuate nutrient decomposition by influencing the composition and biomass of the soil microbiome [121]. Compost soil is a well-known alternative to

synthetic fertilizers, decreasing agricultural waste. Bio-composting is a microbial oxidative activity that leverages reusable organic compounds for plant conservation and growth development in organic agriculture. In numerous investigations, compost and agro-waste have been identified as abundant biological control soil microbe resources. Green compost can subsequently be exploited as a management tool to prevent soil-borne plant diseases. The methodologies and procedures matching naturally suppressive soils have also been implemented in greenhouse-based horticulture soil-less technologies [122]. Farmers lack access to soil fungicide, and fumigants consider compost a dependable resource for disease prevention. Anaerobic soil disinfestation (ASD), which uses wheat bran or ethanol soil supplements as a carbon source, decreased the intensity of root rot infection on tomatoes due to *Pyrenochaeta lycopersici* and transformed soil microbes, enhancing the number of Firmicutes [123].

6.2. Synthetic Microbial Consortia

Synthetic microbial consortia (SMC) are soil microbiota groups similar to synthetic biology because they can reconstruct the shape and components of the plant microbes. It is conceivable to synthesize SMC having various functionalities for plant growth advancement. This seems to have the potential to help solve many of the problems with conventional microbial bio-fertilizers, such as host intolerance, poor competition with indigenous microbes, and inadaptability to the surrounding ecology [124]. The several procedures engaged in designing the proper SMC require choosing the origination of the microbes, extracting and developing the core microbes, improving the microbial interconnections depending on suitability, and evaluating the effectiveness of such a consortium [125]. A bacterial consortium of *Pseudomonas putida*, *Comamonas testosteroni*, *Citrobacter freundii*, and *Enterobacter cloacae* has been discovered to increase phosphate movement and farming practices near twofold [126].

Furthermore, a relationship among the diazotrophic N-fixing bacteria *Azotobacter vinelandii* and the mycorrhizal fungus, *Rhizophagus irregularis*, promoted root development in the field, resulting in increased nutrient absorption in wheat [127]. While the rhizobium microbiome may influence plant development via secreting specific phytohormones [128], recent research developed two synthesized microbiomes consisting of bacterial strains with ACC deaminase activities. Immunization with such synthesized microbial consortium exhibited antimicrobial activity towards *F. oxysporum* f. sp. *lycopersici* decreased the tomato plant symptoms and improved productivity on a weak substrate [129,130].

The present review observed that several concerns remain, and it is questionable how these can be completed via modern technology. The composition of soil amendments, plant breeding/genetic modifications, and focused microbiome engineering can minimize the demand for synthetic fertilizers or pesticides, delivering excellent yields and more resilient crops. The adoption of novel technologies must be focused on existing research for the specific crop, soil, and environment and whether such approaches may be implemented adequately within the conventional farming systems.

6.3. Handling and Inoculation of Microbiome to Plants

The challenge is to formulate a policy for the storage of microbiome such that its functionality is safe, efficient, and well-preserved during application and handling [131]. Almost all developing countries, particularly those in Africa and Latin America, and several Asian countries, do not adhere to laws or norms for installation, and plant beneficial soil microbiota are often produced utilizing non-sterile transporters [132]. Although regulations and recommendations still exist in other countries, such as Brazil, Rwanda, Zambia, and Argentina, they are either not incorporated in producing practices or are not followed [133–135]. The inoculation shall immediately maintain its expected capabilities in translating its possibilities in fields. One major cause is the fast reduction in viable cell count, which often falls below the threshold, and thus becomes useless in agricultural areas [136,137]. A desirable inoculation has much more than about 10^3 colonies per gram

of soil. To accomplish such a rising colony count, transporters with properties including perfect water-holding potential, excellent aeration, and the ability to facilitate microbial development and self-life are required. Furthermore, the carrier must be inexpensive, widely used, and easily available. The liquid compositions are of particular broth (10–40%), dispersant (1–5%), suspension component (1–3%), surfactants (3–8%), and carrier's liquid (possibly oil or water or a mixture of both) at 35–65% by weight [138].

Polysaccharide gums are suggested to obtain viscosity to prevent components from settling while applying colloidal clays, starch cellulose, or synthesized polymers. Furthermore, plant probiotics must include bacteria that play beneficial functions in plant development while safe for people and the environment. Microbes for plant probiotics must be recorded using a polyphasic approach that provides for morphological, chemotaxonomic, and genotypic classification [139]. For fungal and bacterial strain characterization, molecular techniques such as 16 S rDNA or internal transcribed spacer (ITS) sequence analysis DNA-DNA hybrid can also be performed [140]. In addition, the American Biological Safety Association (ABSA) introduced compulsory and must-follow safety evaluation guidelines. Before their delivery, field demonstrations and systematic guidance to growers could enable developing awareness and appropriate inoculants. The guidelines will offer an image based on farmers' farmlands and product commercialization for effective technologies installation. This may eventually lead to a significant increase in the usage of bio compositions in farming activities [141,142]. Because of the absence of standardized protocols for beneficial microbe applications, current regulations and recommendations are usually strictly controlled to rhizobial inoculants diverge between areas. These rules indicate the lower cell count for inoculated per unit of inoculum weight and allow the most contaminants in the end products [139,143]. In a few countries, regulations for the marketing of bio compositions involve rigorous regulatory procedures, and the microbiological compounds are evaluated for such requirements by the sovereign laboratories [142,144].

The first and foremost significant and compulsory suggestions for utilizing beneficial plant rhizobia include assuring that the biocontrol agents are sustainable for the environment. Furthermore, inoculum processing must be from clean cultures and aseptic carriers, containing 10^6 active cells per seed and no contamination. The present study suggests that the use of successful experiment-based methods engaging farmers, the significance of social science research, and the marketing of beneficial soil microbiota utilizing information and communications techniques (ICT) can undoubtedly contribute to the rising popularity of these microbes. Concerns about the sustainability of agroecosystems require the progress of innovation and procedures which have no deleterious influence on the environment and are conveniently available to growers.

Notwithstanding the considerable advancement in agricultural output within the recent fifty years, it might be optimistic to anticipate it will perform proportionately across the period. Management of the plant microbiome is generally limited to specific microbes. However, introducing any isolated strain to soil microbiota is frequently ineffective in promoting plant development and decreasing stress resistance due to interaction with native soil microbiota communities and limited colonization capability.

7. Microbial Pesticides

The researchers have demonstrated that chemical pesticides can easily cause soil compaction, resulting in the death of micro-organisms in the soil, environmental pollution, a series of food safety problems, and severe harm to human health; agricultural growers gradually reduced the use of chemical pesticides [145,146]. Proper use of pesticides in modern agrarian production is the critical means to guarantee the quality of agricultural output, but chemical pesticide residues in agricultural products and the environment will leave many adverse effects on the quality of farm produce and the ecological environment compared with chemical and biological pesticides. Biological pesticides are safer and one of their key characteristics is that they are environmentally friendly, and they also have

been given more attention over the years. Extensive achievements have been made in the research and application of their correlation [147–149].

The preparations made by micro-organisms or their metabolites used as pesticides are called microbiological pesticides, which usually consist of living-cell bacteria as the main composition [150,151]. Agricultural antibiotics produced by micro-organisms are also generally included in the category of microbial pesticides (Figure 4) [152]. According to the microbial resources used, microbial pesticides can be classified into bacteria, fungi, viruses, etc. [153,154]. Their functions can also be classified into insecticides, fungicides, herbicides, and plant growth regulators. Microbial strains used as pesticides are often used to inhibit or kill plant pathogens, pests, or weeds and are often referred to as biocontrol bacteria (Figure 3) [155–157]. They are used primarily as microbial pesticides bacteria, for example, those belonging to the *Bacillus*, *Streptomyces*, *Pseudomonas* genus, etc. [158]. For example, the bacillus, *Escherichia coli*, can be used to prevent and treat the sheath blight rice disease [159,160]; radiation *Agrobacterium* K84 is commonly used to prevent and control plant crown gall disease [161]; some of *Pseudomonas aeruginosa* can be used to produce pyocyanin of bean shell ball spore [162,163]; and soil-borne pathogens have a potent inhibition [17].

Agricultural antibiotics are organic compounds produced by micro-organisms that can inhibit or kill plant diseases caused by bacteria (Figure 4) [164]. Several micro-organisms can be used to produce agricultural antibiotics [165]. The agricultural antibiotics produced by Actinomycetes are the most common, such as streptomycin, hygromycin, jinggangmycin, oxytetracycline, etc. [166–170]. In addition, some agricultural antibiotics also have other functions, such as insecticide or plant growth regulation (Figure 4).

7.1. Application of Reporter Genes

The reporter gene works based on encoding efficiently and frequently detectable signals that indicate the existence or exposure to particular determinants [171]. As a result, micro-organism cells are often outfitted with accurately engineered reporter genes to exhibit the synthesis of unique biomolecules, affecting gene expression [172,173]. Reporter gene products within the influence of a specified regulating system enhance sensitivities using a convenient and efficient sensor system [171]. Enzymes have been widely exploited as reporter gene outcome identifiers to determine pollutant compounds with colorimetric, fluorescent, or luminescent readouts. The frequently implemented enzymes for this objective are galactosidase b-gal, which encodes through the lacZ gene and can identify chemicals immediately and based on colorimetric or fluorescence characteristics [174]. The availability of chemiluminescent and electrochemical substrates for b-gal enables increased sensitivities, as they have an extensive and dynamic sensing area. Fluorescent proteins have also been regularly used during microbial biosensing as reporters require chemical compounds to be included according to their fluorescence intensity abilities (Figure 5). For example, green fluorescent protein (GFP), expressed via the GFP gene, was applied to detect toxins through emitting light, which could be noticed by employing a conventional potentiometer with minimal or no damage to the hosting mechanism [175,176].

Moreover, GFP was used as a fusing tag and pH indicator during environmental contamination studies, and it functioned as a reporter gene in the biosensing process. To put it simply, toxic substances within the ecosystem could cause stress in living cells by interrupting metabolic equilibrium and various physiological systems [177]. The process-sensitive green fluorescent protein, also known as the roGFP modified version, has been developed to monitor the cell's redox state to discover the actual condition. As a result, the reporter gene (roGFP2) was overexpressed in *Escherichia coli* and immobilized on the κ -carrageenan matrix, resulting in a highly functional but sensitive biosensor capable of detecting oxidative stress compounds in a limited period [177].

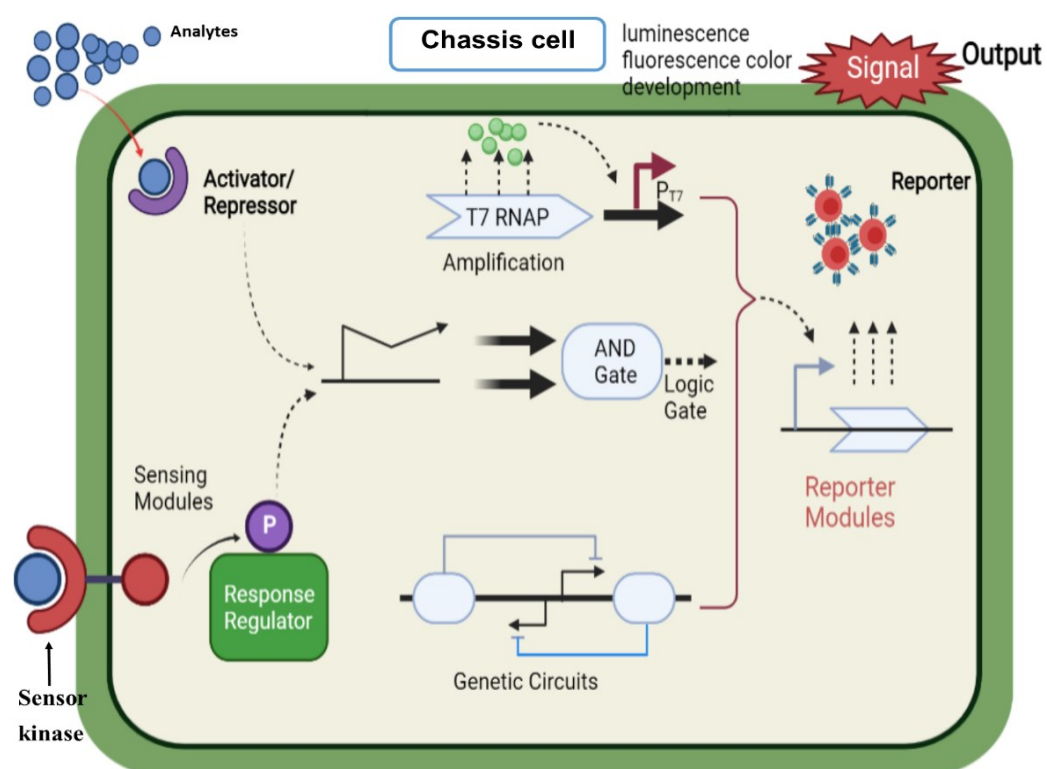


Figure 5. Transference of sensing signals to reporters through the genetic circuits in the whole-cell biosensors.

7.2. Microbial Sensors and Gene Promoter

The promoters are the gene section required to trigger the reporting gene by reflecting the host's existing metabolism. The relevant promoters sections should be preferred during biosensing development, depending on the targeted molecules to be detected [178]. The reporter system's 5'-segment seems to be expected in which a particular promoter gene has been installed. It activates the express reporters whenever it detects the targeted chemical. The selectivity of promoters must be addressed while acquiring compounds [172]. A large percentage of supporters interact with the collection of chemicals rather than a single molecule. This may act in various microbes at periods. Several advocates apply substrate-dependent and host-specific sensing strategies to manage the tracking of a particular function. In the last few years, promoters that have been developed have been characterized by using advanced technologies. For example, metal-induced promoters sections were discovered and combined in cassettes which may be applied to enable reporter mechanisms such as the lux or GFP reporter's gene or the production of readily detectable external membranes epitopes "Figure 5" [177]. Lead and cadmium concentrations have been identified by using the high selectivity of these stimulated gene expressions.

Several different kinds and suitable promoters have been applied to develop pesticide biosensors [179,180]. There are also promoters, such as enzymes, nucleic acids, whole cells, antibodies, aptamers, MIPs, etc., accessible for determining the total toxicity [181,182]. The major disadvantage of developing microbial biosensors is the scarcity of influential promoters that react exclusively to relevant contaminants. To solve this issue, a better understanding of gene regulatory systems in microbes is required. Applying microarray technologies to associate metagenome information plus metatranscriptome studies of microbial colonies might provide a vast demand for additional regulatory components in the upcoming years [183]. Another alternative may be to develop "super promoters" focusing on consensus sequencing derived through comparative research on various promoters within identified regulatory systems [184].

7.3. Advancements in Biosensors through Nanotechnology Application

Obstructions in micro-organisms sensors' functionality include challenges with this transducer's immobilization process, size inappropriateness, sensitivity, and specificity. Nanotechnologies with tiny and broad surface areas, guaranteeing enhanced surface activity and electrical conductivity, are becoming more appealing and efficient technologies. Nanomaterials, nanostructures, nanocomposites, and fiber optics are commonly exploited in the development of microbial biosensing [185]. Carbon nanotubes are appropriate for the immobility of microbial biosensing according to their electrical behavior and large outward zone. They can develop beneficial electro-chemicals that perform better and improve stability [186]. Furthermore, to these capabilities, carbon nanotubes help enhance the cell's loading activities, hestin, catalytic performance, and electrical conductance [187].

Nano-particles (NP), particularly gold (Au), are based on high conductance, biocompatibility, and catalytic properties and are being used to modify the surface morphology of electrodes to derive maximum immobilized efficiency [188,189]. They can facilitate the electron mobility between the microbe cell walls and the electrode surface, thus protecting the natural state of redox-active proteins. In addition, a Au NP-advanced conduction of polymer expressed excellent biocompatibility, stability, sensitivities, and the capability for electro-catalysis while applied as a medium for the determination of glucose [189].

Fiber optic-based technologies have been developed to immobilize micro-organisms for quick identification of analysts. Due to their rapid reaction and robust immobility properties, fiber optic-based biotechnologies have an opportunity over conventional biosensors [190]. A real-time biosensor for detecting contamination in freshwater was developed by applying a flowing through fiber optic-based flowing sensors [191]. Because the bacteria are immobilized on the fiber optic biosensor, the biosensor offers regulating advantages, even as bacteria cannot depart the detector into the water [190,191].

According to previously published articles, such an approach's success involves a specific strain's capability to perform multiplex-sensing to various analytes in a material through building the cells arrays (Figure 5). Moreover, nanotechnology has facilitated the verification of microbial biosensors via increasing cell immobilization, localization, mobility, and stability characteristics. Still, a comparative study is required to apply chitosan derived from the various microbes and their application as pesticides with biosensing capabilities [192].

8. Application of Microbes in Pesticide Degradation

Microbes (fungal and bacterial) applications have been identified for the bio-degradation of agro-chemicals within the environment [193]. The efficiency of microbial species to bio-degrade chemicals varies considerably [194]. Pesticide remediation using microbes transforms harmful chemicals into nontoxic, eco-friendly, and beneficial metabolites [195]. During pesticides decomposition, the biosorption rate for a single strain is insufficient [196,197], whereas the focus of degradation studies is rapidly turning towards microbiological consortiums, and pesticide bio-degradability is determined through pesticide components, available mechanisms, and the promiscuity of enzymes (Figure 3). Certain pesticides break down relatively faster than others [194]. The slower ones are trinitrotoluene (TNT), polychlorinated biphenyl (PCBs), and pentachlorophenol (PCP). However, methomyl, pyrethroids, 1,3-dichloropropene, and atrazine can degrade faster [198–200]. Axenic cultured cells are concentrated more on pesticides breakdown than microbial consortia [201]. Earlier research has examined various microbial communities that, particularly axenic strains, may degrade chemicals quickly. During investigations, both single and a mixture of microbial strains are effective. Although axenic cells seem critical in metabolic studies, their physiology and molecular compositions are related to pesticide decomposition. The synthesis of the consortium was achieved premised on the performances of axenic colonies in pesticide degrading, and the microbial consortia became identified to have the tremendous potential [202,203].

8.1. Microbial Biosensors and Applications in Sensing Pesticide Residues

Farmers use different chemical fertilizers and pesticides to enhance crop productivity [204,205]. Certain agro-chemicals are intentionally utilized in soil fertilizing to prevent infestations, bacterial and fungus infection, grasses, nematodes, and rodents [206]. These agricultural chemicals remnants ultimately infect the ecosystem and nutrition supply, potentially direct or indirect. The continuous introduction of agrochemicals into the environment raises residue buildup and impacts living organisms, particularly humans [207,208]. The most important groups of chemicals include organophosphate, organo-nitrate, organochlorine, and related compounds, which are destructive to different living micro-organisms in the ecosystem. To provide a comprehensive awareness of the existing concentrations of pesticides, one must look at residue accumulation and the contaminants sustained, interacting pathways with the soil, and the biota observed in a particular region [209,210].

The microbial sensor is one of the emerging applications for assessing pollutants by coupling micro-organisms with a transducer to allow analysts to detect from various sources in a fast, precise, and sensitive manner [211]. Previously, microbe biosensors strongly depended on functional cell respiratory and metabolic activities to recognize a chemical, potentially a substrate, or inhibit their metabolic mechanisms [212]. Moreover, existing microbial biosensors are constituted of transmitters that interact with fixed viable or non-viable microbial cells, exceptionally genetically engineered species. Non-viable micro-organisms targeted periplasmic enzymes identified in permeability cells or whole-cells were evidenced to be cheaper than cellular enzymes. The transportable/portable cell ensembles of biosensing were also developed using freeze-dried biosensors isolates of micro-organisms for high-throughput pollution assessment [213]. Although microbes are highly effective at exploiting a broader range of chemical compounds according to their diverse metabolic culture, genetically modified capabilities, and resistance to an extensive range of ecological factors, microbial biosensors are much more beneficial for research on pesticides in the coming years.

However, several researchers have introduced advanced, sensitive, dependable, and productive chromatographic technologies to determine chemical compounds from the samples of environmental materials. Furthermore, these techniques are time-consuming and labor-intensive, and they require complex and expensive instruments and competent specialists [214]. These challenges and problems should be solved where biosensors are the cheapest and most excellent alternative for chemical analysis. Therefore, these bio-reporters (antibodies, whole-cell, DNA, enzymes, and RNA) have been applied to the development of biosensing and have already been revealed to be practical components [215]. Certain elements are widely modified through the action and execute particular functions. However, microbial biosensors are whole-cell with biological reporter genes, and many are linked with physio-chemical sensing to provide signal processing [216]. Signal production may be accomplished through variations in proton level, gas release or uptake, and luminescence, depending on the nature of the microbe's metabolic process of the compounds.

8.2. Microbes in Biochemical Degradation of Pesticides

The primary motive behind mixing the cultures' sustained functionality is to minimize metabolic stress and labor allocation throughout pesticide degradation. However, single-step microbiological biotransformation has delivered more effectiveness in agro-chemical degrading, particular pesticides with different substrates could require multiple systems and degradation activities because they can be decomposed using a single strain [217]. All those complicated activities may be fulfilled by a cohort of different micro-organisms, individually engineered for targeted functionalities that facilitate collaboration at the macro level of the population [218]. This is found in the natural ecosystem, wherein single strains need not function effectively, and many species provide supportive responsibilities [198]. Contaminants can be degraded by microbes exploiting specific metabolic systems. Pesticides decomposition genes, mRNA, enzymatic, and metabolism, could be applied to analyze the entire systems biology in microbes [217]. Different bio-informatics strategies

are developed to recognize every micro-organism's gene regulatory systems and cell biology throughout pesticide breakdown. Systems biology research of individual microbe strains and colonies in the laboratories and fields may describe the regulatory procedures and environmental limitations driving microbial ecosystem performances. Integrating innovative microbial information obtained from systems research findings within current nutrients chain tools will improve overall predicting capabilities, which is crucial for forming mitigating approaches and regulations in modern pesticide conditions [198,219].

8.3. Pesticide Degradation through Microbial Engineering

Agro-chemicals can be degraded through microbial consortia present across the environment. While chemicals enter soil and groundwater sources, a microbial consortium uses its capabilities to fix them via various metabolic mechanisms [220]. Microbial consortia are more tolerant to environmental variations in ecosystems than monocultures; they potentially execute complex functions [221]. Quorum Sensing Promotes the allocation of pesticide degrading work, allowing complicated activities to be completed. A consortium may include genera that are related or unrelated, as well as a diversity of species. The consortium components interact via various procedures, including neutralism, cooperation, amensalism, competition, predation, commensalism, and microbial strains exhibiting neutralism cannot engage with others, enabling them to degrade agro-chemicals independently [222]. In contrast, each strain can degrade in commensalism, but the other does not. The relationship that emerges whenever a nearby strain's development affects the metabolic products of one strain is characterized as amensalism.

These strains produce alternative primary metabolites when competing for the same chemical pesticides. During predation, components of the secondary strain impede the development of those from the initial. The cooperating process enables the strains to benefit from each other, although the initial strain (A) metabolites may sometimes limit themselves throughout pesticide breakdown. Syntrophic relationships in chemical degradation are enabled through microbial strain collaboration. Two bacterial species, *Sphingomonas* sp. TFEE and *Burkholderia* sp. MN1 have been found to decompose the pesticide fenitrothion using syntrophic interactions, whereas individual microbes in a consortium have diverse metabolic pathways that can execute complicated tasks effectively [223–225]. Several auxotrophic strains cultivated simultaneously and have compatible metabolic activities may sustain each other in a syntrophic co-culture [226,227]. Connectivity within the natural microbial community usually involves transmitting diffusible signal molecules among different microbes. The majority of bacterium QS is mediated through acyl-homoserine lactone (AHL), the diffusible signal component in Gram-negative bacteria. It can play a role of an arbitrator via peptides in Gram-positive bacteria [228,229].

Quorum Sensing underpins the fundamental ideas of communication and operating effectiveness in a pesticide degrading consortia [230]. Engineered microbial interactions are the first process in forming a synthesized consortium. Gene-level modification of QS molecules is critical to the efficacy of the structuring of a synthesized microbial population. Combined microbial strains emerge whenever a few species localize and collaborate to provide a particular function [231]. However, a few of the collaboration's microbial strains do not engage in metabolic activities, but are promoted by the metabolism of additional consortium participants. These strains are usually referred to as cheating strains [232]. To engineer extensive and accessible consortia, several genome-wide and computational methodologies could be implemented. An engineering model system will help with large-scale pesticide decomposition inside the environment [233]. Computational capabilities are required to understand the behaviors and consequences of modified metabolic processes and anticipate the communities' dynamics. Chemical degrading mechanisms can be determined using in silico technologies [234].

These types of technologies could be helpful in anticipating and developing novel metabolic interactions between engineered microbial communities. Engineering sustainable microbial communities for pesticide degradation seems important; hence, strategies for

long-lasting and strong microbiomes for pesticide bioremediation in the ecosystem should be developed [213]. Our review suggests that developing an artificial microbial consortium under natural conditions requires an accurate assessment of its development, output, and activity. The biocontainment of such microbial consortiums could guarantee that the augmented features will not damage natural ecosystems.

9. Conclusions and Future Prospectus

While writing this paper, we felt a great apprehension about the significance of our living environment and a heightened responsibility for the future of these delicate resources becoming imperiled by the adverse effects of climate change. Plant microbiome and associated beneficial microbes, which we may often “take for granted” and even fail to notice, are very fragile and, at present, are depleted consistently. Though beyond the scope of this review, we can state that due to injudicious crop/pest management practices, we are on the course of losing many microbe species, which will ultimately imbalance the microbial ecology for future generations. Microbes are essential to plant health and fitness components for determining crop production and disease tolerance. For sustainable and clean food production, deploying microbes is a tremendously smart approach to the non-transgenic and communally protracted plant genome. Since some microbes can contribute to plant health and yield, they are considered to have a bright future as they are cost-effective, pollution-free, and sustainable agriculture that covers elsewhere more naturally distinct plant–microbe interactions. Understanding microbes’ evolution from isolated individuals to an organized community interrelating with plants provides opportunities with a considerable impression in organic agriculture. Microbe-based bioproducts potentially characterize a sustainable and active approach to reducing global climate change’s biotic/abiotic stress.

Furthermore, microbes can contribute to the balance of the agro-ecosystem and environmental health, diminishing insecticides, agro-chemicals, and fertilizers in extensive farming. However, in the persuasion of improved production efficiency and extra pervasive employment, some questions must be well-thought-out in cooperation with regulators and at the research and development level. Biotic and agro-ecological research and microbes still present numerous restrictions mostly related to lower efficiency and complex environmental sensitivity compared to other agro-chemicals. Additionally, microbial products sometimes exhibit unpredictable outcomes for different crops or regions. Consequently, to take full advantage of the effective use of microbial products, future research should be encouraged to obtain improved target-based products, for example, by deep exploration of plant microbiomes.

Bio-fertilizers can be coupled with agronomic practices to escalate agro-ecosystem biodiversity and safeguard a long and constant synergetic association with crop plants. In this situation, the utilization of microbial products characterizes a justified and effective way out, though there may be crop production losses estimated for climate change. It may perhaps help to augment human contributions to the agriculture ecosystem.

There is a crucial need to increase efforts toward the ramifications of climate change on important food and cereal crops, using this information to initiate predictions of climate effects, in due course, to propose microbiological approaches to combat global warming and environmental degradation. Even though we reviewed some techniques for harnessing plant-microbiome to help alleviate climate change and global warming effects, in no way do we recommend that it will be adequate to counterpoise the microbial losses that are already occurring. As an alternative, integrated management is immediately desirable, which may require the best possible practices for harnessing resilient microbes, conservation of biodiversity, carbon sequestration, and sustainable smart agriculture. For the achievement of these goals, it is necessary to conduct more studies related to microbiome; biotechnological research ought to be used to encourage experimental or demonstrative trials for all the appropriate interested parties, from farmers to agriculture extension services and policymakers, to confirm the upfront application of microbes in various crops, areas, and

environments. Mutual understanding, collaboration, and persistent data exchange among researchers, stakeholders, and policymakers bring a fruitful validation of the scientific approach in actual conditions and adaptability in practical applications. Most prominently, political will is essential as well as a globally steadfast determination to restrain the emission of greenhouse gases, which are the primary cause of global warming. We notice that these problems connected with the existing “caution to people” may help as a premonition for the prominence of microbes to establish a bio-network sustainability and climate-resilient agriculture during global warming.

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